

PU-I Biology · Chapter 6

Anatomy of Flowering Plants

Handwritten Notes — Narendra Sir · BAIO Learning System

1 The Tissue System

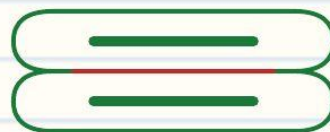
Def ▶ Tissues are grouped into **3 tissue systems** on the basis of structure & location: **epidermal**, **ground (fundamental)**, **vascular (conducting)**.

Epidermal tissue system

- Epidermis — single layer of parenchyma, **cuticle** (waxy) outside — absent in roots
- **Stomata**: 2 guard cells enclosing a pore + **subsidiary cells** = stomatal apparatus · regulate transpiration & gaseous exchange ★ exam
- Guard cells have chloroplasts; outer walls thin, inner walls (towards pore) thick



Dicot — bean-shaped guard cells



Grasses — dumb-bell shaped

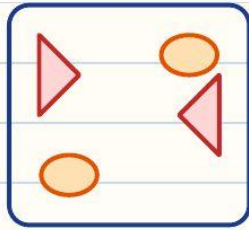
- Epidermal appendages: **root hairs** (unicellular, from epidermis) · **trichomes** (stem, multicellular, prevent water loss)

Ground tissue system

- All tissues except epidermis & vascular bundles — parenchyma, collenchyma, sclerenchyma
- Stems/roots: hypodermis, cortex, pericycle, pith, medullary rays · Leaves: **mesophyll**

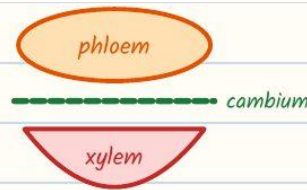
Vascular tissue system

Xylem + phloem = vascular bundles. Cambium between them → **open**; no cambium → **closed**.



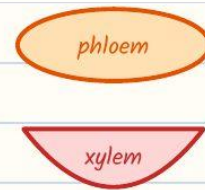
Radial

X & P on different radii · roots



Conjoint open

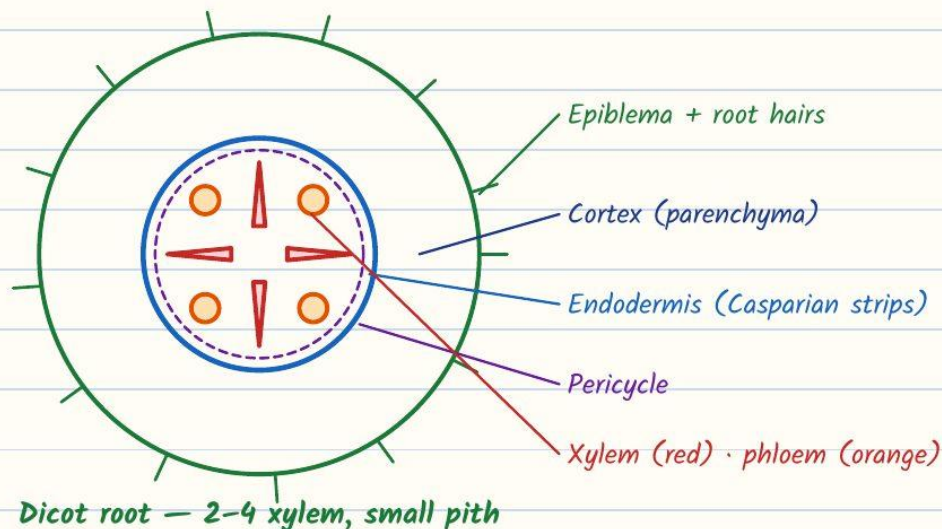
cambium present · dicot stem



Conjoint closed

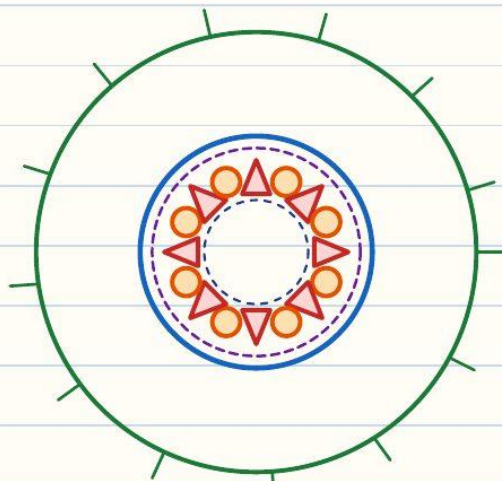
no cambium · monocot stem, leaves

2) Dicot Root



- **Epiblema** — outermost layer, many cells bear unicellular root hairs
- **Cortex** — several layers of thin-walled parenchyma with intercellular spaces
- **Endodermis** — innermost cortex, barrel-shaped cells, no intercellular spaces ·
Casprian strips of suberin (water-impermeable) ★ **exam**
- **Stele** = pericycle + vascular bundles + pith (inside endodermis)
- **Pericycle** — thick-walled parenchyma · initiates **lateral roots** and vascular cambium (secondary growth)
- Vascular bundle **radial**, xylem **exarch** · 2-4 xylem & phloem patches · pith small/inconspicuous
- **Conjunctive tissue** — parenchyma between xylem and phloem

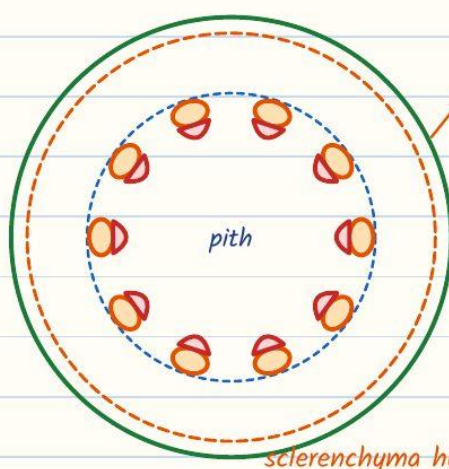
3 Monocot Root



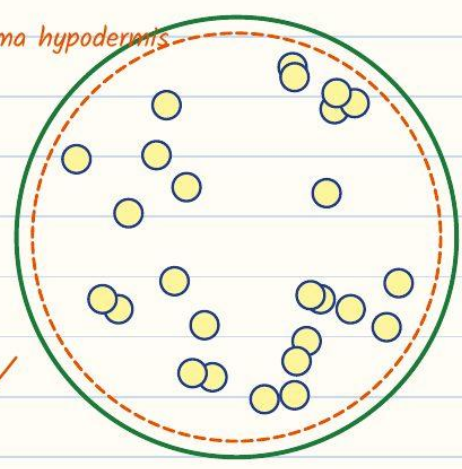
Monocot root — polyarch, large pith

- Same tissues: epiblema, cortex, endodermis, pericycle, vascular bundles, pith
- Vascular bundles radial, exarch, **polyarch** — usually **more than 6** xylem bundles
- Pith large & well developed
- No secondary growth — no cambium formed ★ exam

4 Dicot & Monocot Stem



Dicot stem — VBs in a ring, pith



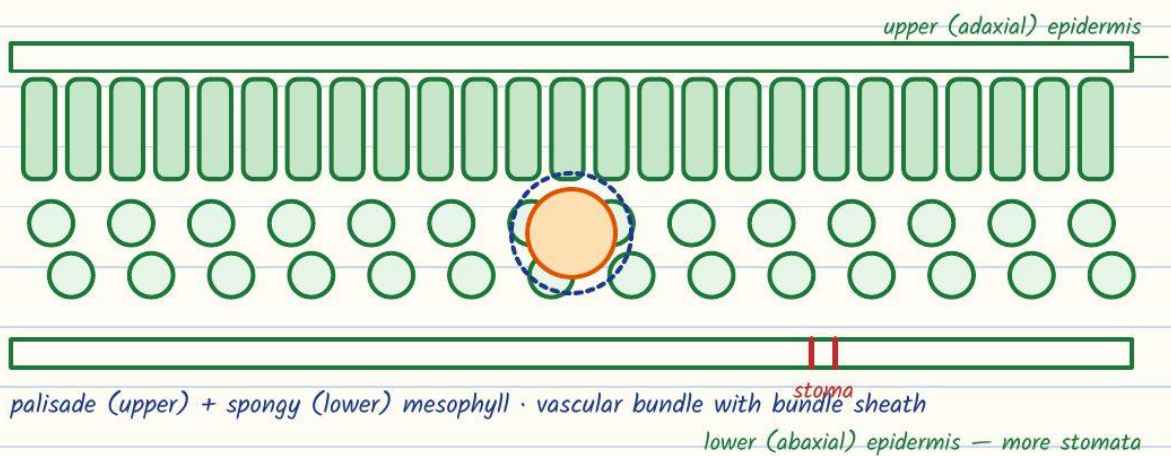
Monocot stem — scattered VBs, no pith

Dicot stem (sunflower)

- Epidermis with cuticle, may have trichomes & few stomata
- **Hypodermis = collenchyma** (mechanical strength to young stem)
- Cortical layers: parenchyma · innermost = **endodermis** rich in starch → **starch sheath**
- Pericycle = semilunar patches of **sclerenchyma**
- VBs in a **ring** — conjoint, **open**, endarch protoxylem
- Medullary rays between VBs · large pith

Monocot stem (maize)

- **Hypodermis = sclerenchyma**
- Large, conspicuous parenchymatous ground tissue
- VBs **scattered**, each with a sclerenchymatous **bundle sheath**
- VBs **conjoint, closed** (no cambium)
- Peripheral VBs smaller than central ones
- **Phloem parenchyma absent** · **water-containing cavities** in VBs

5 Dicot & Monocot Leaf**T.S. of a dorsiventral (dicot) leaf**

- **Dicot leaf — dorsiventral**: upper (adaxial) & lower (abaxial) epidermis with cuticle; **abaxial has more stomata** (adaxial may lack them)
- Mesophyll = **palisade** (adaxial, elongated, vertical, parallel) + **spongy** (abaxial, oval/round, loose, large air spaces)
- Vascular bundles in veins & midrib; size depends on vein size · surrounded by thick-walled **bundle sheath** cells · reticulate venation
- **Monocot leaf — isobilateral**: stomata on **both surfaces**, mesophyll **not differentiated** into palisade & spongy

- **Bulliform cells** — large, empty, colourless cells on adaxial epidermis of grasses ·
 turgid → leaf open; flaccid (water stress) → leaf rolls inward to reduce water loss
- ★ exam
- Monocot veins nearly same size (except main veins) → parallel venation

6 Dicot vs Monocot – Comparison

Feature	Dicot	Monocot
Root xylem	2–4 (di- to tetrarch)	> 6 (polyarch)
Root pith	small / absent	large
Stem hypodermis	collenchyma	sclerenchyma
Stem VBs	ring, open	scattered, closed, bundle sheath
Stem pith	present	absent (ground tissue)
Leaf	dorsiventral · palisade + spongy	isobilateral · undifferentiated mesophyll
Stomata	more on lower surface · bean-shaped	equal on both · dumb-bell (grasses)
Secondary growth	yes (cambium)	no

💡 **Remember: Golden rules:** radial + exarch = root · conjoint + endarch = stem ·
 open = dicot · closed = monocot · bulliform cells = grass leaf.

Quick revision ✓

Stomatal apparatus = guard + subsidiary cells · Casparian strips = suberin in endodermis · pericycle → lateral roots · dicot stem: collenchyma hypodermis, starch sheath, ring of open VBs · monocot stem: scattered closed VBs, bundle sheath, water cavities · bulliform cells → leaf rolling